James McAllister

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TECHNICAL SUMMARY

Software Engineering student bridging AI/ML and full-stack development. Experienced in shipping production mobile applications with integrated deep learning systems, building scalable ML APIs, and implementing end-to-end AI features from model training to deployment. Combines strong software engineering fundamentals with practical AI/ML expertise

EDUCATION

Bachelor of Science in Software Engineering *(ABET Accredited)* May 2028
Minors: Computer Science, Mathematics, Data Science
Shippensburg University of Pennsylvania; Wood Honors College
GPA: 3.56 Major GPA: 3.76

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, Dart, C, HTML/CSS, SQL, NoSQL
Frameworks and Libraries: React, Flutter, FastAPI, PyTorch, TensorFlow, NumPy, Pandas, ChromaDB
AI/ML: Deep Learning, Computer Vision, CNNs, Transfer Learning, Vector Databases, Inference optimization
Developer Tools: Git, GitHub, Firebase, REST APIs, OpenAI API, USDA API, Figma, Kaggle, Google Cloud
Concepts: Object-Oriented Programming, Data Structures, Algorithms, Agile Development, Cloud Architecture

PROJECTS

Prodigy (In Development) | *Solo AI Engineer* December 2025 - Present
- Engineering full-stack AI assistant integrating OpenAI API, ChromaDB vector database, Google Calendar/Tasks APIs with React frontend and FastAPI backend
- Architecting RAG pipeline with document parsing, variable chunking, embedding generation, and semantic retrieval for context-aware Q&A with source citations
- Building an OAuth2 authentication system for secure Google service integration with automatic token refresh and session management across 15+ API endpoints
- Implementing a 3-layer memory architecture managing session state, persistent knowledge base, and conversational context
- Building a responsive React interface with real-time chat, document management, task boards, calendar integration, and settings

Tech Stack: Python, FastAPI, React, OpenAI API, ChromaDB, Google Calendar/Tasks API, PDF processing, TypeScript

LifeLens | *Lead Designer and Engineer* *August 2025 - December 2025*
- Developed a deep learning computer vision system using CNNs and transfer learning (ResNet architecture) to classify cancer images with 90% validation accuracy across 20,000+ medical images from Kaggle datasets
- Architected a RESTful API backend with FastAPI to serve ML models and handle multi-modal data processing, integrating structured medical data with image classification results
- Built a responsive React frontend with OpenAI API integration for natural language explanations of risk assessments and medical disclaimers
- Applied Grad-CAM visualization techniques for model explainability and interpretability

Tech Stack: Python, TensorFlow, PyTorch, NumPy, Pandas, FastAPI, React, TypeScript, OpenAI API, Kaggle, Git

BeneFit | Founder and Lead Software Engineer June 2025-September 2025
- Engineered and deployed a full-stack cross-platform mobile application using the Flutter framework for iOS and Android, currently in closed beta testing
- Designed and implemented a scalable NoSQL database architecture using Firebase Firestore with real-time synchronization and Firebase Authentication for secure user management
- Integrated USDA API with 300,000+ food items searchable in real-time, macro/calorie tracking with data visualization using custom chart components and a progress analytics dashboard, and interactive displays/charts
- Developed a gamification engine with XP-based leveling, 13+ challenges, an achievement system, step tracking features and social features (friend streaks, rankings) to drive engagement
- **Tech Stack**: Flutter, Dart, Firebase (Auth, Firestore, Cloud Functions), REST APIs, Figma, Git

EXTRACURRICULAR ACTIVITIES

- Shippensburg Club Lacrosse Team *September 2024 - Present*
- Programming Team *September 2025 - Present*